

This document provides additional assistance with wiring your Extron IPL PRO enabled product to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instruction, refer to the user's manual for the specific Extron IPL PRO enabled product or the controlled device manufacturer supplied documentation.

Device Specifications:

Device Type: Video Conference
Manufacturer: Cisco
Firmware Version: TC7.3.1
Model(s): C90 TC7.3.X, C60 TC7.3.X, C40 TC7.3.X, C20 TC7.3.X

Tested on the Following Software and Firmware Versions:

IP Link Pro Control Processor Firmware	Touch Link Panel Firmware	GS Version
2.03.0000-b006	2.01.0001-b001	1.1.1

Version History

Module Version	Date	Notes
1_0_0_1	10/21/2016	Updated Module. Updated Phonebook.
1_0_0_0	4/19/2016	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'. Example: InterfaceName.Unidirectional = 'True'.
- connectionCounter variable must be set the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15. Example: InterfaceName.connectionCounter = 5.
- Driver will only work with firmware TC7.3.4 or older.
- devicePassword and deviceUsername variables must be set accordingly. Example: InterfaceName.devicePassword = 'admin'. InterfaceName.deviceUsername = 'TANGBERG'.
- CallHistoryOccurrence specify if Call History is ordered based on time of occurrence or frequency of occurrences is used to set how many entry in the call history to display. The default value is 'Time'. It must be set to 'Time' or 'Frequency'. Example: InterfaceName.CallHistoryOccurrence = 'Frequency'.
- CallHistoryNumber is used to set how many entry in the call history to display. The default value is 5. It must be set to a number between 1 to '20. Example: InterfaceName.CallHistoryNumber = 10.
- PhoneBookSearchNumber is used to set how many results are returned from the search. The default value is 5. It must be set to a number between 1 to 10. Example: InterfaceName.PhoneBookSearchNumber = 10

Phonebook commands:

- PhonebookSearchResult
 - This is where the search results will be saved to when phonebook is searched. The number of results are set by a driver parameter Number of Phonebook Search, this is initialized to 5 results by default. Simply drag this onto a Text Feedback object for displaying the search result.
- PhonebookSearchSet
 - When the number selected in this command is triggered, it will copy the name in the corresponding Phonebook Search Result Button over to the Dial String command for dialing.
- PhonebookNavigation
 - This command is used for navigating through the search results, it is like scrolling through the results when there are more than the Number of Phonebook Searches that came back from the search.
- PhonebookUpdate
 - This is used to trigger the search again.
- PhonebookFolderID
 - This command is used to configure a Folder ID dynamically and is sent out of the port by using Phonebook Update.
- PhonebookFolderIDUpdate
 - This is used to trigger the search again.
- PhonebookFolderIDSearchSet
 - When the Folder Name is selected this command is triggered, it will copy the Folder Name in the corresponding Phonebook Folder ID Search Result Button over to the Phonebook Folder ID command.
- PhonebookFolderIDSearchResult
 - This is where the search results will be saved to when Phonebook Folder ID is searched. The number of results are set to 20 results. Simply set this onto a Text Feedback object for displaying the Phonebook Folder ID Search Result.

Control Commands, States, and Qualifiers:

AssignLocalOutput ¹	None
['Layout ID']	1 to 2147483647 in steps of 1
['Output ID']	0 to 65534 in steps of 1
AudioOutput	'On' 'Off'
['Output']	'Line 1' – 'Line 6' ¹ 'Line 1' – 'Line 2' ^{2,3}
AutoAnswer	'On' 'Off'
CallHistoryNavigation	'Up' 'Down' 'Page Up' 'Page Down'
CallHistoryRefresh	None
CallHistorySelect	None
['Button']	'1' – '20'
CallSetupMode	'Gatekeeper' 'Direct'
CameraFocus	'Near' 'Far' 'Stop' 'Auto'
['Camera']	'1' – '7' ^{1,2,3}
CameraFocusC20	'Near' 'Far' 'Stop' 'Auto'
CameraPan	'Left' 'Right' 'Stop'
['Camera']	'1' – '7' ^{1,2,3}
['Speed']	1 to 15 in steps of 1
CameraPanC20	'Left' 'Right' 'Stop'
['Speed']	1 to 15 in steps of 1
CameraPresetPosition Recall	'1' – '15'
['Camera']	'1' – '7' ^{1,2,3}
CameraPresetPosition RecallC20	'1' – '15'
CameraPresetRecall	'1' – '35'
CameraPresetSave	'1' – '35'
['Camera']	'1' – '7' ^{1,2,3}
CameraPresetSaveC20	'1' – '35'
CameraSerialNumber Refresh	None
CameraTilt	'Up' 'Down' 'Stop'

['Camera']	'1' – '7' ^{1, 2, 3}		
['Speed']	1 to 15 in steps of 1		
CameraTiltC20	'Up'	'Down'	'Stop'
['Speed']	1 to 15 in steps of 1		
CameraZoom	'In'	'Out'	'Stop'
['Camera']	'1' – '7' ^{1, 2, 3}		
['Speed']	1 to 15 in steps of 1		
CameraZoomC20	'In'	'Out'	'Stop'
['Speed']	1 to 15 in steps of 1		
CloseMenu	None		
DisplayMode	'Auto'	'Single'	'Dual'
	'Dual Presentation Only'	'Triple Presentation Only' ^{1, 2}	'Triple' ^{1, 2}
	'Quadruple' ¹		
DNSDomainNameCommand	'String'		
DNSServerAddressCommand	'String'		
DoNotDisturb	'Active'	'Inactive'	
DTMF	'0' – '9'	'*'	'#'
FarEndCameraPan/Tilt	'Up'	'Down'	'Left'
	'Right'	'Stop'	
FarEndCameraPresetRecall	'1' – '15'		
FarEndCameraPresetSave	'1' – '15'		
FarEndCameraSource	'0' – '15'		
FarEndCameraZoom	'In'	'Out'	'Stop'
FarEndControl	'On'	'Off'	
H323GatekeeperAddresCom mand	'String'		
H323AliasE164Command	'String'		
H323AliasIDCommand	'String'		
H323ProfileAuthenticationLo ginNameCommand	'String'		
H323ProfileAuthenticationPa sswordCommand	'String'		
Hook	'Dial'	'Disconnect All'	'Join 1'
	'Accept'	'Reject'	'Hold 1'
	'Resume 1'	'Disconnect 1'	'Hold 2'

	'Hold 3'	'Resume 2'	'Resume 3'
	'Disconnect 2'	'Disconnect 3'	'Join 2'
	'Join 3'	'Hold All'	'Join 4'
	'Join 5'	'Hold 4'	'Hold 5'
	'Disconnect 4'	'Disconnect 5'	'Resume 4'
	'Resume 5'		
['Protocol']	'H320'	'H323'	'SIP'
	'Auto'		
['Number']	'String'		
Input	'1' – '5' ¹	'1' – '3' ^{2, 3}	'1' – '2' ⁴
InputLineLevel	0 to 24 in steps of 1		
['Input']	'Line 1' – 'Line 4' ¹	'Line 1' – 'Line 2' ²	
InputMicLevel	0 to 70 in steps of 1		
['Input']	'Microphone 1' - 'Microphone 8' ¹	'Microphone 1' - 'Microphone 4' ²	'Microphone 1' - 'Microphone 2' ³
InputMute	'On'	'Off'	
['Input']	'Microphone 1' - 'Microphone 8', 'Line 1' - 'Line 4' ¹	'Microphone 1' - 'Microphone 4', 'Line 1' - 'Line 2' ²	'Microphone 1' - 'Microphone 2', 'Line 1' - 'Line 2' ³
IPv4AddressCommand	'String'		
IPv4GatewayCommand	'String'		
IPv4SubnetMaskCommand	'String'		
IREmulation	'0' – '9'	'Delete'	'Call'
	'Disconnect'	'F1'	'F2'
	'F3'	'F4'	'F5'
	'Grab'	'Home'	'Layout'
	'Mute'	'Mute Mic'	'Ok'
	'Phonebook'	'Presentation'	'Selfview'
	'Aux'	'Camera'	'Doc Camera'
	'PC'	'VCR'	'Volume Up'
	'Volume Down'	'Zoom In'	'Zoom Out'
	'*'	'#'	'Up'
	'Down'	'Left'	'Right'
Layout	'Add'	'Remove'	'Remove All'
	'Reset'		
['Layout ID']	1 to 65535		
LayoutSet	'Auto'	'Custom'	'Equal'

Global Scripter Module
Communication Sheet

	'Fullscreen'	'Overlay'	'Presentation Large Speaker'
	'Presentation Small Speaker'	'Prominent'	'Single'
	'Speaker Full'		
['Target']	'Local'	'Remote'	
MicrophonesMute	'On'	'Off'	
MissedCallNotification	'On'	'Off'	
Output Level	-24 to 0 in steps of 1		
['Output']	'Line 1' – 'Line 6' ¹	'Line 1' – 'Line 2' ^{2, 3}	
PhonebookFolderID	'String'		
PhonebookFolderIDNavigation	'Up'	'Down'	'Page Up'
	'Page Down'		
['Phonebook Type']	'Corporate'	'Local'	
PhonebookFolderIDSearchSet	1 – 10		
PhonebookFolderIDUpdate	None		
['Phonebook Type']	'Corporate'	'Local'	
PhonebookNavigation	'Up'	'Down'	'Page Up'
	'Page Down'		
['Phonebook Type']	'Corporate'	'Local'	
PhonebookSearchSet	1 – 10		
PhonebookUpdate	'Refresh'	'Next Search'	'Previous Search'
['Phonebook Type']	'Corporate'	'Local'	
['Contact']	‘String’		
['FolderID']	‘String’		
PictureInPicture	'On'	'Off'	
Presentation	'1' – ‘5’ ¹	'1' – ‘3’ ^{2, 3}	'1' – ‘2’ ⁴
	'Stop'		
PresentationSendingMode	'Local and Remote'	'Local Only'	
PresetRecall	'1' – '15'		
PresetSave	'1' – '15'		
Reboot	None		
Self View	'On'	'Off'	

SelfViewDefaultFullscreenMode	'On'	'Off'	'Current'
SelfViewPosition	'Upper Left' 'Center Left' 'Lower Right'	'Upper Center' 'Center Right'	'Upper Right' 'Lower Left'
SleepTimer	1 – 480		
SNCameraFocus	'Near' 'Auto'	'Far'	'Stop'
['Serial Number']	'String'		
SNCameraPan	'Left'	'Right'	'Stop'
['Serial Number']	'String'		
['Speed']	1 to 15 in steps of 1		
SNCameraPresetSave	'1' – '35'		
['Serial Number']	'String'		
SNCameraTilt	'Up'	'Down'	'Stop'
['Serial Number']	'String'		
['Speed']	1 to 15 in steps of 1		
SNCameraZoom	'In'	'Out'	'Stop'
['Serial Number']	'String'		
['Speed']	1 to 15 in steps of 1		
SpeakerTrackControl	'On'	'Off'	
SpeakerTrackMode	'Auto'	'Off'	
Standby	'Activate'	'Deactivate'	
Volume	0 to 100 in steps of 1		

¹ Only available on C90

² Only available on C60

³ Only available on C40

⁴ Only available on C20

Status Available:

AudioOutput	'On'	'Off'	
['Output']	'Line 1' – 'Line 6' ¹	'Line 1' – 'Line 2' ^{2, 3}	
AutoAnswer	'On'	'Off'	
CallHistory	'String'		
['Button']	'1' – '20'		
['Detail Type']	'Display Name'	'Callback Number'	'Last Occurrence Time'
	'Occurrence Count'	'Occurrence Type'	
CallStatus	'Dialing'	'Connecting'	'Connected'
	'Ringing'	'Idle'	'Disconnecting'
	'On Hold'	'Early Media'	'Preserved'
	'Remote Preserved'		
['Call']	'1' – '5'		
CallStatusType	'Video'	'Audio'	'Audio Can Escalate'
	'Forward All Call'	'Unknown'	
['Call']	'1' – '5'		
ConnectionStatus	'Connected'	'Disconnected'	
DisplayMode	'Auto'	'Single'	'Dual'
	'Dual Presentation Only'	'Triple Presentation Only' ^{1,2}	'Triple' ^{1,2}
	'Quadruple' ¹		
DisplayName	'String'		
['Call']	'1' – '5'		
DNSDomainName	'String'		
DNS ServerAddress	'String'		
['Server']	'1' – '3'		
DoNotDisturb	'Active'	'Inactive'	
FarEndControl	'On'	'Off'	
FirmwareVersion	'String'		
GatewayAddress	'String'		
H323GatekeeperAddress	'String'		
H323GatekeeperStatus	'Required'	'Discovering'	'Discovered'
	'Authenticating'	'Authenticated'	'Registering'
	'Registered'	'Inactive'	'Rejected'
Input	1 – 5 ¹	1 – 3 ^{2, 3}	1 – 2 ⁴

InputLineLevel	0 to 24 in steps of 1		
['Input']	'Line 1' – 'Line 4' ¹	'Line 1' – 'Line 2' ^{2, 3}	
InputMicLevel	0 to 70 in steps of 1		
['Input']	'Microphone 1' – 'Microphone 2'		
['Input']	'Microphone 1' - 'Microphone 8' ¹	'Microphone 1' - 'Microphone 4' ²	'Microphone 1' - 'Microphone 2' ³
InputMute C90	'On' 'Off'		
['Input']	'Microphone 1' – 'Microphone 8'		'Line 1' – 'Line 4'
	'Microphone 1' - 'Microphone 8', 'Line 1' - 'Line 4' ¹	'Microphone 1' - 'Microphone 4', 'Line 1' - 'Line 2' ¹	'Microphone 1' - 'Microphone 2', 'Line 1' - 'Line 2' ¹
IPAddress	'String'		
MACAddress	'String'		
MicrophonesMute	'On' 'Off'		
MissedCallNotification	'On' 'Off'		
NetworkAssignment	'Static' 'DHCP'		
OutputLevel C90	-24 to 0 in steps of 1		
['Output']	'Line 1' – 'Line 6' ¹	'Line 1' – 'Line 2' ^{2, 3}	
Presentation	'1' – '5' ¹ 'Stop'	'1' – '3' ^{2, 3}	'1' – '2' ⁴
Presentation Mode	'Off'	'Sending'	'Receiving'
PresentationSendingMode	'Local and Remote'	'Local Only'	'Off'
RemoteNumber	'String'		
['Call']	'1' – '5'		
SelfView	'On' 'Off'		
SelfViewDefaultFullscreenMode	'On'	'Off'	'Current'
SelfViewPosition	'Upper Left' 'Center Left' 'Lower Right'	'Upper Center' 'Center Right'	'Upper Right' 'Lower Left'
SIPRegistrationStatus	'Deregistering' 'Registered'	'Failed' 'Registering'	'Inactive'
SpeakerTrackControl	'On' 'Off'		
Standby	'Activate' 'Deactivate'		
SubnetMask	'String'		
Volume	0 to 100 in steps of 1		

¹ Only available on C90

² Only available on C60

³ Only available on C40

⁴ Only available on C20

Cable and Adapter Requirements:

C20: Captive Screw to USB Type-A RS-232 Serial Cable

C40, C60, C90: Captive Screw to Male DB9 RS-232 Serial Cable

Notes for the Device:

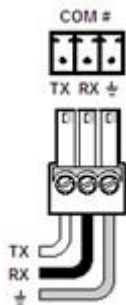
Serial communication:

Port Type: RS-232
Baud Rate: 115200
Data Bits: 8

Parity: None
Stop Bits: One
Flow Control: None

Pin Assignments Diagram

C20:

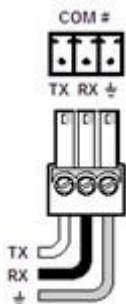


Signal	Main Cable	Pin	Signal
RxD	←	2	TxD
TxD	→	3	RxD
GND		4	GND

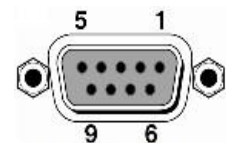


Video Conference

C40, C60, C90:



Signal	Main Cable	Pin	Signal
TxD	↗	2	TxD
RxD	↖	3	RxD
GND		5	GND



RS232 DB9, Female
Video Conference

General Notes:

Network communication:

When configuring the Ethernet driver, be sure device settings match that of the GCP configuration.

Port Type:	Ethernet
Default Port:	23
Logon Credentials Supported:	Yes
Default User Name:	admin
Default Password:	TANDBERG
Multi-Connection Capabilities:	Yes
Port Changeability:	No

Ethernet Module Configuration Description:

Please refer to user manual for settings and changes to the network communication

Notes for the Device:

Telnet mode must be set to 'On' by an admin in order to have open communication with the driver.